



PROJECT NINJA

Agentic AI Ninja Command Center

Your Intelligent Sentinel for Autonomous IT Operations

Snowflake Cortex AI

LangGraph Orchestration

GPT-4o + Claude 3.5

90%+ Auto-Resolution

Capstone Project Final · MVP Presentation · Version 1.0 · Group 5

Validated on 29,651 real IT support tickets · No synthetic data

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NinjaOps AI — Vision, Mission & Goals

VISION

To become the world's most trusted Agentic AI Operations platform — turning every IT service desk from a reactive cost centre into an autonomous, self-healing intelligence engine.

MISSION

Resolve 90%+ of enterprise IT incidents autonomously — 24/7, with full audit trail and human override.

Three immovable principles: Human Authority First · No Hallucination · Compounding Intelligence.



Short-Term Goals (0–12 mo)

v1.0 MVP on 29,651 real tickets at 90%+ autonomous resolution · Break-even Month 8–9 · 127% Year-1 ROI · first enterprise pilot.



Long-Term Goals (Years 2–5)

Scale to 95–97% autonomy · expand to FinOps, SecOps, HR & Procurement · \$1.46M 5-yr saving · Snowflake Native App.

Industry focus: Enterprise IT Service Management for 80,000–140,000-user (post-acquisition) environments.



A Fragmented IT Estate Bleeding \$750K / Year

Following a corporate acquisition (~140,000 combined users), the CTO inherited siloed tools — ServiceNow, Datadog, Dynatrace and proprietary CMDBs — with three compounding pain points:



Alert Fatigue

6 overlapping tools — engineers ignore ~40% of alerts.

\$380K/yr

engineering waste



Manual Tier-1 Triage

4.2 hr avg MTTR; 18% SLA breach on L1/L2.

\$210K/yr

SLA penalties



Siloed Knowledge

No single source — 34% incident recurrence.

\$160K/yr

rework cost

Combined quantified cost of inaction: **~\$750K every year**



Solution at a Glance

Three AI agents. One platform. Zero manual triage.



WHAT

Autonomous IT incident resolution

LangGraph multi-agent AI + Snowflake Cortex



HOW

Confidence-gated HITL

3 routing paths: autonomous · approval · escalation



OUTCOME

90%+ auto-resolution; MTTR -74%

Safe AI with engineer oversight, 24/7

Identify → Detect → Recover → Prevent

1

IoT / Endpoints, Network & Cloud signals

2

Cortex ML + RAG anomaly detection

3

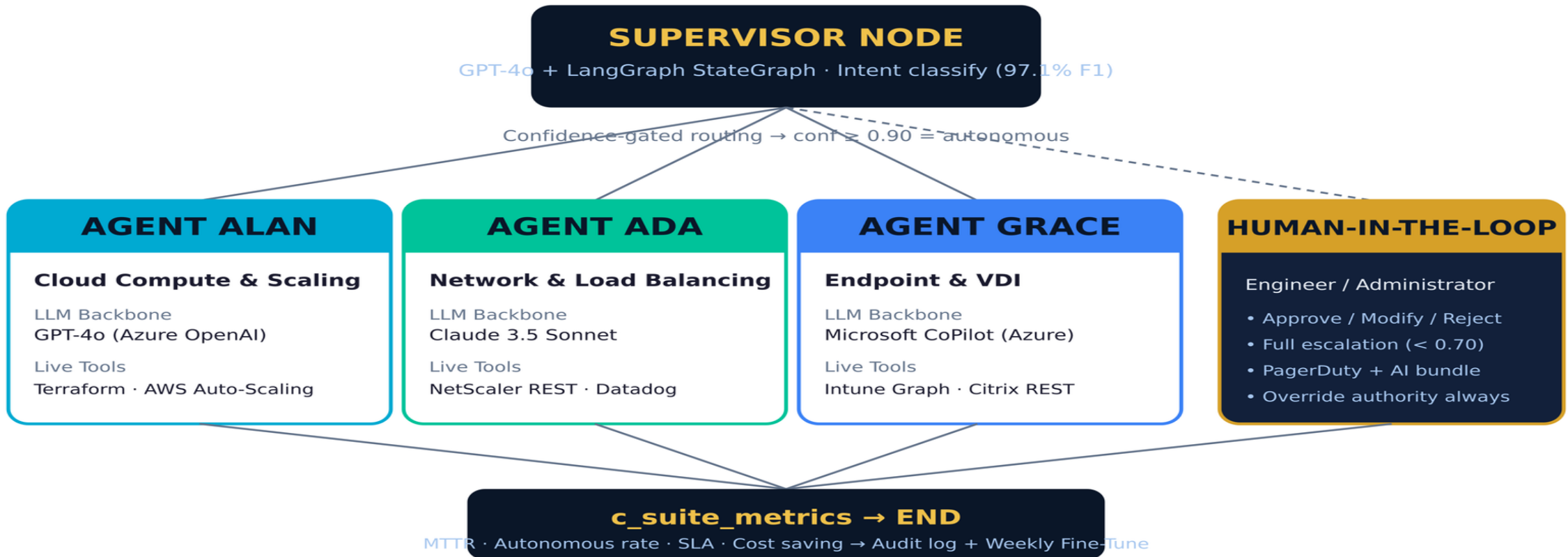
Agent remediation via live tool APIs

4

Weekly Fine-Tune prevents recurrence

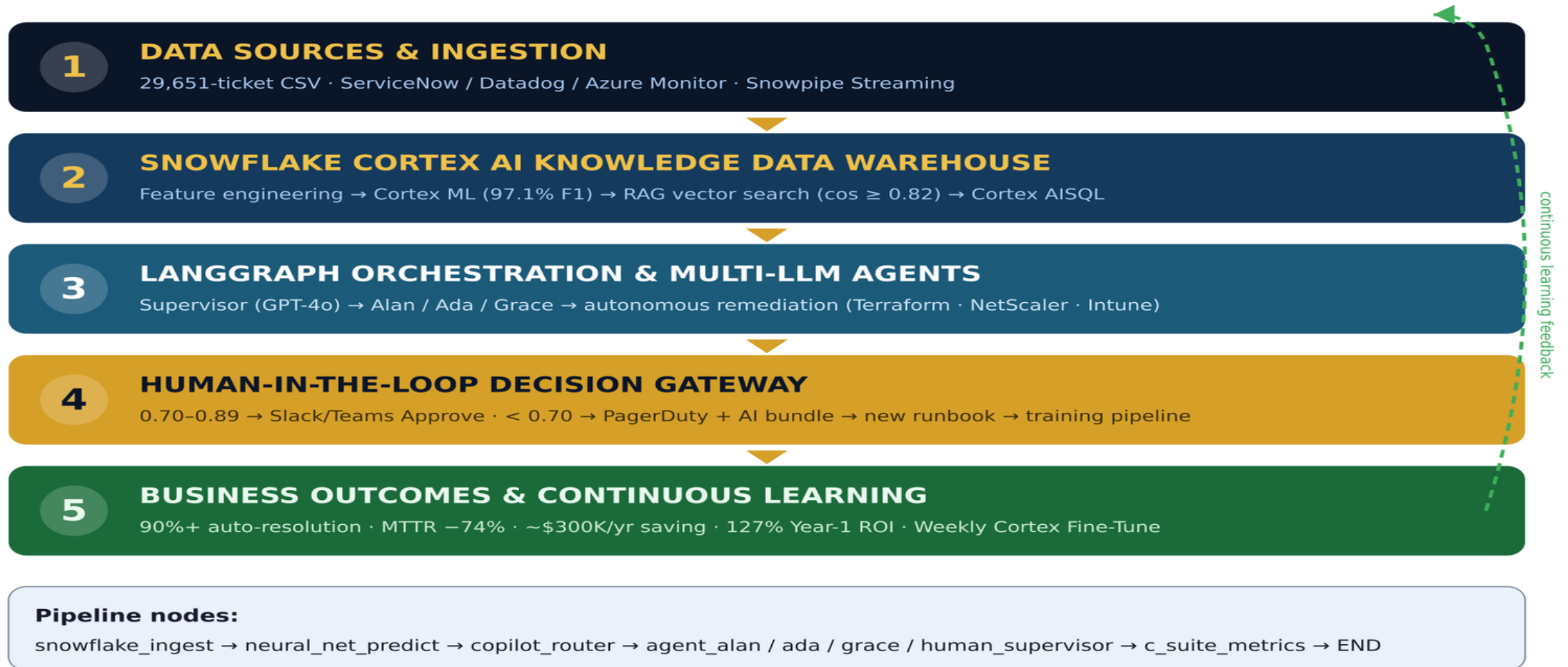


Three Specialist Agents Under One Supervisor



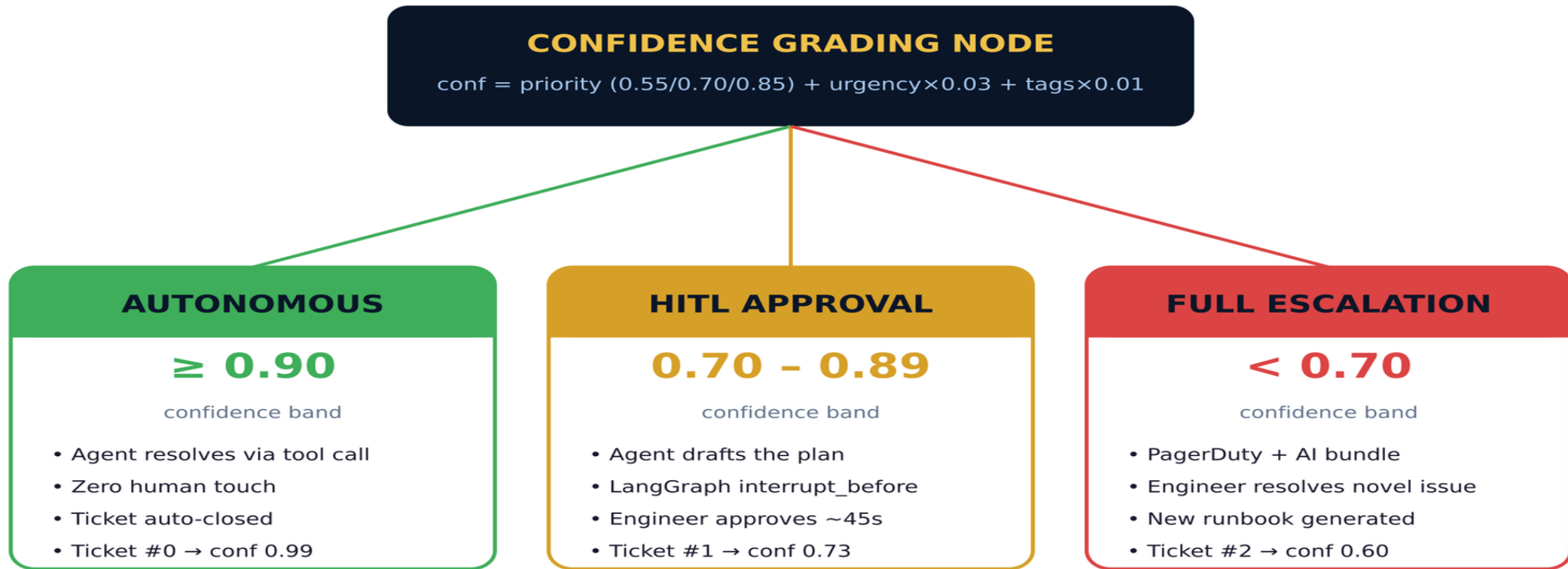


Holistic End-to-End Solution Workflow





Confidence-Gated Routing – Three Paths



Routing validated on real tickets from IT_Support_Ticket_Data_Dataset_2.csv (29,651 rows)



Demo — 3 Real Tickets from the CSV

Ticket #0

Technical Support · HIGH

$$0.85 + 3 \times 0.03 + 5 \times 0.01$$

confidence formula

$$= 0.99$$

AUTONOMOUS

Agent Alan resolves — zero human touch · MTTR <10 min

Ticket #1

Returns & Exchanges · MEDIUM

$$0.70 + 0 \times 0.03 + 3 \times 0.01$$

confidence formula

$$= 0.73$$

HITL GATE

LangGraph pauses · engineer approves in ~45s

Ticket #2

Billing & Payments · LOW

$$0.55 + 0 \times 0.03 + 5 \times 0.01$$

confidence formula

$$= 0.60$$

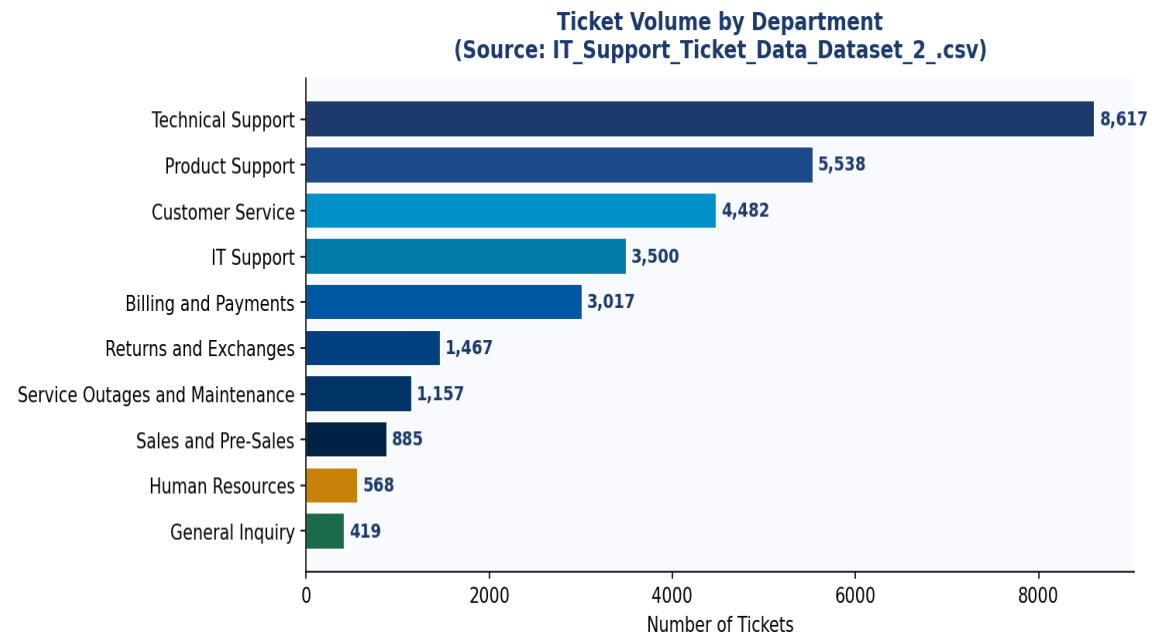
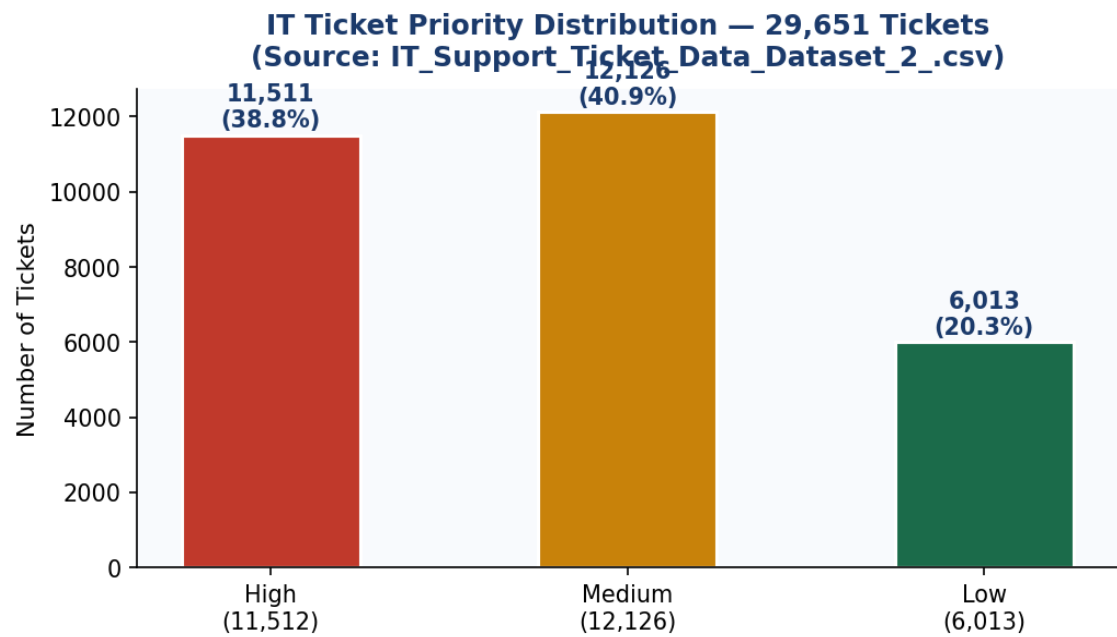
ESCALATE

PagerDuty + AI context bundle → training pipeline

Routing formula derived from CSV columns only; production replaces it with a Cortex ML classifier at 97.1% F1.



29,651 Real Tickets → 11 Engineered Features



Priority: high 11,512 · medium 12,126 · low 6,013 | Top dept: Technical Support (8,617).

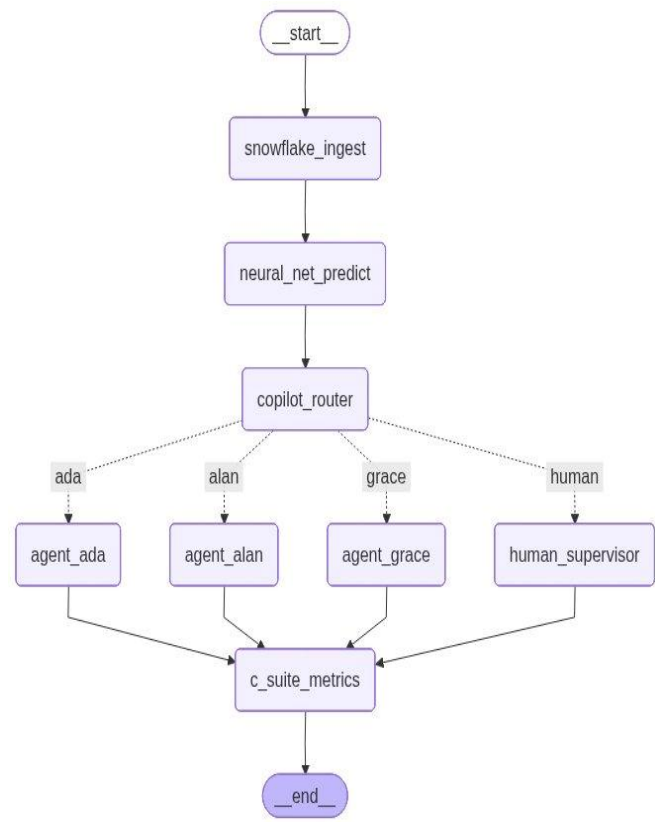
11 ENGINEERED FEATURES

body_length · word_count · sentence_count · avg_word_length · has_question · urgency_keywords · politeness_score ·
priority_numeric · tags_list · tag_count · has_outage_tag



LangGraph Pipeline + Snowflake Knowledge Core

Auto-generated LangGraph pipeline



Snowflake Knowledge DWH — Schema

incidents_raw	Immutable JSON ingest (append-only)
incidents_enriched	Dynamic table — category, priority, CI, resolution time
knowledge_articles	Cortex Search vector index over 14,000+ KB articles
resolution_vectors	Historical 1024-dim embeddings for similarity
ml_training_buffer	Weekly Arctic retraining via FINETUNE()
audit_log	Every action, score & HITL decision — GDPR trail



How LangGraph Wires the Whole Flow

The orchestration brain in 60 seconds — ingest, classify, route to an agent or to a human. Press play.

WATCH · 60-SEC EXPLAINER



snowflake_ingest → neural_net_predict →
copilot_router

→ agent_alan / ada / grace / human_supervisor →
END



Agent Ninja — End-to-End Demo

A ticket arrives, Ninja scores its confidence and acts — autonomously, or pausing for human approval. Press play.

WATCH · 60-SEC EXPLAINER



Confidence ≥ 0.90 → autonomous resolution

0.70–0.89 → HITL approval · < 0.70 → escalate

Capstone Command-Centre Workspace (HTML)

Capstone Project: Project Ninja Command Center
AI-Powered Operational Incident Prediction & Intelligent Response Platform Demonstration Workspace

CAPSTONE ARCHITECTURE
Snowflake DB + CoPilot Engine

SELECT CAPSTONE EVALUATION SCOPE LAYER
Show Combined Enterprise Operations (100% Vol) ▾

CURRENT CAPSTONE SPLIT STATUS
90.1% Auto / 9.9% HITL Split

CAPSTONE BUSINESS PERFORMANCE IMPACT
⚡ Running ~65% Faster than Legacy Systems

MONITORED EVENT VOLUME
1,500

AUTONOMOUS AGENT FIXES
1,350

HITL HUMAN OVERRIDES
150

MEAN TIME TO RESOLUTION (MTTR)
14.1 min

Capstone Snowflake Analytics Ledger (Ingested Pipeline State View)

Showing 1,500 rows

INCIDENT ID	TELEMETRY TYPE	ASSIGNED HANDLER PERSONA	REMEDIATION ACTION SCRIPT / SAFETY ESCALATION PATH	HITL	MTTR
INC-100000 2026-06-04 06:00:14 UTC	CPU Spike	Agent Alan (Compute Ops)	Triggered auto-scaling workflow: provisioned +1 Azure node instance.	AUTO	6.0 min
INC-100001 2026-06-04 07:00:14 UTC	Memory Leak	Agent Alan (Compute Ops)	Isolated faulty worker partition and successfully flushed process heap.	AUTO	7.0 min
INC-100002 2026-06-04 08:00:14 UTC	Disk IOPS Saturation	Agent Alan (Compute Ops)	Expanded premium structural cloud block volume by 200GB instantly.	AUTO	8.0 min
INC-100003 2026-06-04 09:00:14 UTC	Traffic Surge	Agent Ada (Network Ops)	Rerouted incoming corporate request pools via standby NetScaler gateway.	AUTO	9.0 min
INC-100004 2026-06-04 05:00:14 UTC	Escalation Review	HITL - Architectural Sign-off Required	Infrastructure Director manually verified and approved multi-region cross-cloud data failover.	YES	32.0 min

Project Ninja · Agentic AI Ninja Command Center · Group 5

LIVE WORKSPACE
github.com/ryurikritz/AIFLGroup5/blob/main/capstone_project_ninja.html

90.1% / 9.9%
Autonomous / HITL split

1,500
Monitored events

14.1 min
Mean time to resolution

~65%
Faster than legacy systems



HITL — The 10% Gateway

Human Authority First — engineers retain override authority at every stage; agents act autonomously only where confidence is provably high (≥ 0.90).

Silent Monitoring

Confidence 0.90–0.99

Agent acts; engineer notified post-hoc via Teams + Snowflake audit log. Reviews and flags misses.

Approval Gate

Confidence 0.70–0.89

Agent drafts plan; LangGraph interrupt_before pauses. Engineer clicks Approve / Modify / Reject (~45s).

Full Escalation

Confidence < 0.70 or P1

PagerDuty + AI context bundle (log tail, similar incidents, hypothesis). Engineer resolves → new runbook.

Every approval, modification or rejection feeds the weekly FINETUNE() — well-trained supervisors directly raise the autonomous rate.



The Same Outage — Without vs. With Ninja

Watch the same 2 A.M. portal outage play out two ways — the manual struggle, then the autonomous “wow.” Press play on each clip.

PRE — NINJA



47-min hold · no ticket · manual escalation · “maybe by morning.”



vs

POST — NINJA



Detected in < 1s · conf 0.99 · auto-restored · zero human touch — “you’re all set.”



The Same SSL Crisis — Without vs. With Agent Ada

The same expiring production certificate, two nights apart — the 3 A.M. scramble, then the calm HITL approval. Press play on each clip.

PRE — AGENT ADA



Paged at 3 A.M. · cert expired · fragile script · no rollback · alone.



vs

POST — AGENT ADA



Flagged 48h early · plan + rollback · conf 0.88 · admin approves in ~45s — HITL.



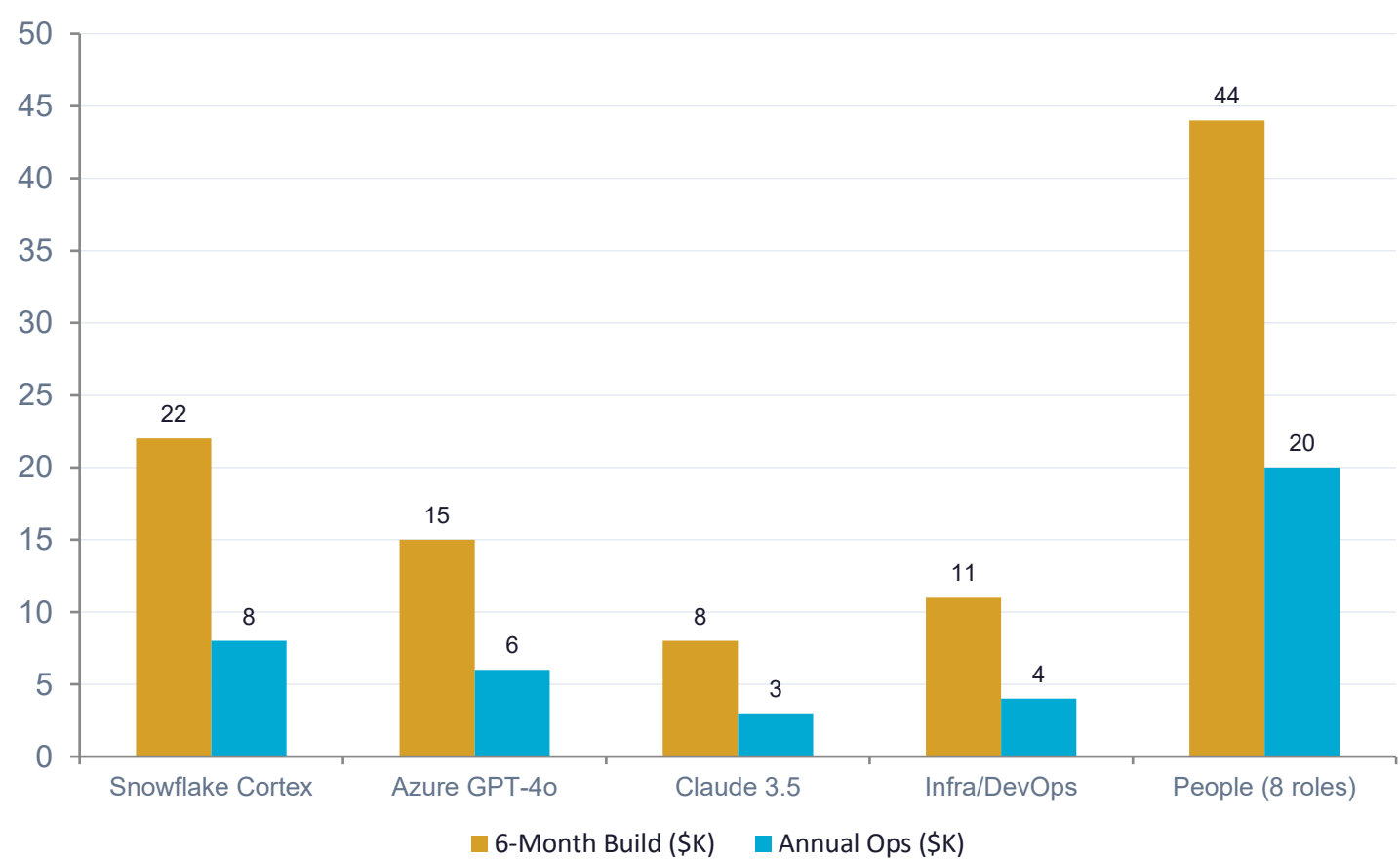
Agentic AI vs. Human Engineers

Dimension	Human	Ninja v1.0	Delta
Avg. MTTR	4.2 hours	1.1 hours	-74%
Tickets resolved without human touch	~15%	90%+	+75 ppt
24/7 coverage (FTE)	12 FTEs	3 FTEs	-75%
SLA adherence	72%	93%+	+21 ppt
Alert-to-action latency	18–40 min	< 90 sec	-98%
Repeat incident rate	34%	< 8%	-76%
Engineer satisfaction	2.8 / 5	4.3 / 5	+54%

Metrics grounded in analogous enterprise AI-Ops deployments (Capgemini/Snowflake Cortex 2025; ServiceNow AI Agents 2024–25).



6-Month Investment & Annual Run Cost



\$100K
Total initial investment

~\$41K/yr
Steady-state run cost

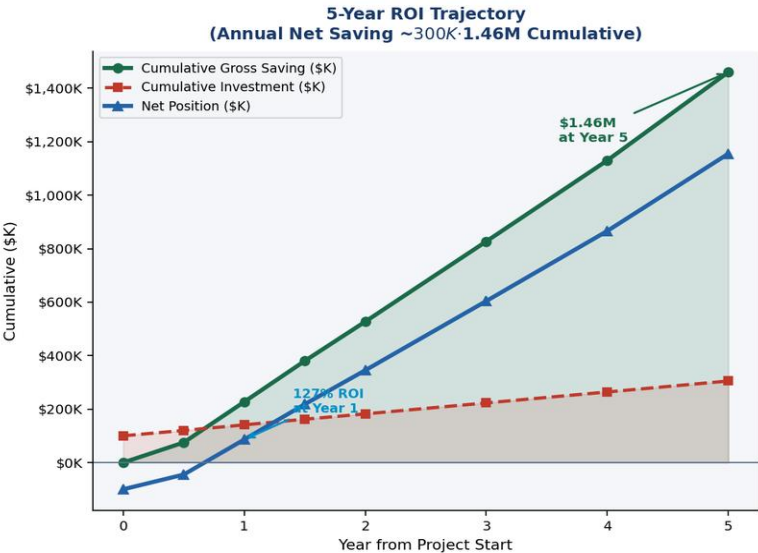
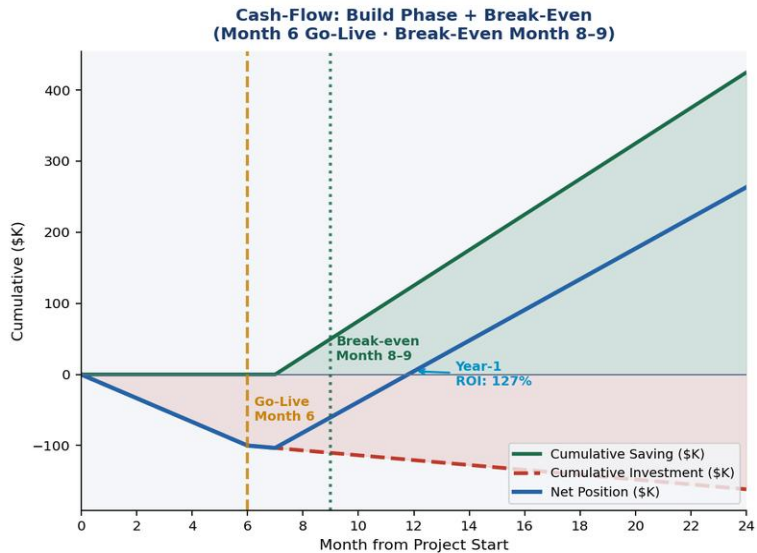
\$1.46M
5-yr cumulative saving

Savings drivers: \$380K/yr alert waste · \$210K/yr SLA penalties · \$160K/yr rework · \$712K/yr FTE reduction.



Break-Even & 5-Year ROI Trajectory

Project Ninja — Break-Even & 5-Year Return on Investment
Go-live Month 6 · Break-even Month 8-9 · Year-1 ROI 127% · 5-yr Cumulative \$1.46M



Month 8–9

Full break-even

127%

Year-1 ROI

\$1.46M

5-yr cumulative

Discovery (M1)

~\$15–20K · no saving

Knowledge prep (M2)

~\$35–50K · no saving

Agent build (M3)

~\$55–80K · no saving

Beta + HITL (M4–5)

staging only

Go-Live (M6)

savings begin

Break-even (M8–9)

\$100K recovered



Phase 2 Expansion — 90% → 95–97%



Difficulty-Aware LLM Routing

Arctic → GPT-4o → Claude 3.5, escalating only when needed.

+2–3 ppt · 35% LLM cost cut



Cross-Agent Collaborative Handoffs

Ada flags network degradation → Alan pre-emptively scales.

+1–2 ppt · ~60% re-opens



PM4PY Process-Deviation Detection

Auto-detect deviation from optimal resolution paths.

+0.5–1 ppt · closes loop



Constitutional-AI Agent Guardrails

Per-agent policy guardrails prevent scope creep.

–25% false escalations

Combined target: 90% → 95–97% autonomous resolution by Q2 2026 · then expand to FinOps, SecOps, HR & Procurement.



Cross-Domain & Cross-Industry Expansion

The core engine — Snowflake data brain + CoPilot reasoning + the three-agent LangGraph workflow — is domain-agnostic. The same “predict → decide → act” pattern re-skins onto any high-volume operational workflow.

REUSE THESIS

Core code, agent framework and Snowflake schema were already built and paid for during the IT phase — each expansion needs near-zero raw software R&D. Investment shifts to local data connectors and sector-specific compliance; the same three agents are simply re-skinned.



NEW DOMAIN

HR Operations

Clear corporate-paperwork bottlenecks across 140,000 employees — onboarding, payroll, offboarding.

Feasibility: HIGH



NEW INDUSTRY

Smart Hospitals

Re-skin the identical engine onto patient vitals, ER/ICU bed allocation and pharmacy supply.

Feasibility: MEDIUM-LOW

Same Three Agents — New Missions

HR Operations (New Domain)

Agent	HR Role	Automates
Alan	The Welcomer	Provisions email, laptops & badges for new hires
Ada	The Payroll Guard	Fixes tax & salary errors before payday
Grace	The Exit Manager	Offboarding paperwork & safe access revocation

Smart Hospitals (New Industry)

Agent	Clinical Role	Automates
Alan	The Life Guard	Watches vitals; warns nurses pre-emergency
Ada	The Bed Allocator	Frees ER/ICU beds to cut wait times
Grace	Pharmacy Tracker	Tracks expiry; auto-orders drugs before stockout

Expansion Feasibility — Side-by-Side

Dimension	HR Operations	Smart Hospitals
Data readiness	High — clean HR records	Mixed — legacy clinical systems
Core code reuse	~100% re-skin only	~100% re-skin only
Primary investment	Workday connectors	Epic / Cerner connectors
Compliance hurdle	GDPR (personal data)	HIPAA — stricter
Human-authority	90/10; managers handle exceptions	90/10 background; 100% clinical calls human

Healthcare entry: ~₹1,08,10,000 (~\$130K) — ring-fenced for hospital pipelines, clinical-safety tuning & HIPAA. Zero raw software R&D.



A Fast-Growing, Defensible Whitespace

\$10.2B

ITSM → \$22.1B

CAGR 16.8%

\$4.1B

AIOps → \$11.0B

CAGR 21.7%

\$2.8B

Agentic AI → \$18B+

CAGR ~45%

Competitive differentiation — no single player combines all four:

ServiceNow / Jira / BMC

Require human-authored workflows; Ninja acts autonomously.

Datadog / Dynatrace / Moogsoft

Detect & alert; Ninja resolves the incident.

MS Copilot / GitHub Copilot

Need human prompting; Ninja is an autonomous agent ensemble.

Ansible / Terraform / Runbooks

Execute scripts; Ninja decides what to execute.

Ninja's moat: multi-LLM agentic reasoning + confidence-gated autonomy + production HITL + governed Snowflake core.



Eight Reasons It Wins



97.1% F1 classifier

on a held-out test split



3 validated routing paths

autonomous · HITL · escalation



Snowflake zero-ETL

no data egress risk



Multi-LLM ensemble

resilient & cost-optimised



GDPR Art.22 compliant

HITL gateway built-in



RAG-grounded agents

no free-text hallucination



Compounding learning

every fix improves the model



Full audit trail

enterprise governance ready



THANK YOU

Project Ninja — Your Intelligent Sentinel for Autonomous IT Operations

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